

Question number	Answer	Additional guidance	Mark
(a)(i)	An answer that includes the following points - C is the correct answer - A is not the correct answer as the iris does not have rod cells - B is not the correct answer as the pupil does not have rod cells - D A is not the correct answer as the spinal cord does not have rod cells		(1)

Question number	Answer	Additional guidance	Mark
(a)(ii)	A description that includes three of the following points: - rhodopsin is the {photoreceptor / photosensitive pigment} (1) - when light absorbed cis retinal (+ opsin) is converted to trans retinal (+ opsin) (1) - causes sodium channels to close (1) - membrane becomes hyperpolarised (1)	accept absorbs light accept Na⁺ gated channels blocked	(3)

Indicative content – description from experimental data

(6)

experiment 2 shows that substance responsible for phototropism passes through gel but not plastic

experiment 1 shows that cause of phototropism is chemical in nature and is produced in the tip
experiment 3 chemical substance can diffuse and cause curvature even in the dark

Chemical nature of responses

phototropism caused by auxins
receptor proteins (phototropins) activate several hormones including auxin
Auxin transported down dark side and less on bright side
Auxin stimulates elongation on shaded side to give curvature

Detailed mechanism of responses

Auxin stimulates decrease in cell pH which activates enzymes responsible for cell growth and elongation
Auxin affects hydrogen ion pumps (proton pumps)
Auxin causes intracellular reserves of hydrogen ions to increase which lowers the pH on dark side
lower pH activate enzymes
these disrupt hydrogen cross links in cell wall
increased osmotic gradient due to movement of solutes
water enters cell (and increased turgor pressure) leads to swelling of cell

Level 1: Discuss some of the results from experimental data.

Level 2: Discuss all of level 1 plus reference to specific chemical nature of phototropism

Level 3: Discuss all of level 2 plus detailed mechanism for phototropism at molecular level